

Abstract No.

Characterization of F3 (Tissue Factor) Expression and Its Role in Angiogenesis Within Glioblastoma Stem Cells

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Background: Glioblastoma (GB) is a highly vascularized brain tumor characterized by intense angiogenesis, heavily driven by Glioblastoma Stem Cells (GSCs). Tissue Factor (F3) is known to be overexpressed in GB, yet its precise mechanistic role in GSC-mediated angiogenesis remains to be fully elucidated.

Objective: This study aims to investigate the biological role of F3 expression in GSCs, evaluate its downstream angiogenic network, and determine its direct influence on endothelial cell angiogenic capacity.

Methods: To dissect the underlying molecular mechanisms, we established an in vitro knockout (KO) model for F3 in GSCs. Transcriptomic profiling was performed to identify altered angiogenic pathways, followed by qPCR and Western blot analyses to specifically validate the expression of key factors, including VEGFA and F3. Additionally, the functional impact on endothelial cells was evaluated using conditioned media from F3-KO and wild-type GSCs, combined with EdU (5-ethynyl-2'-deoxyuridine) proliferation assays to assess endothelial cell turnover.

Results: Transcriptomic data and molecular validation via qPCR and Western blot confirmed the successful depletion of F3 and revealed a significant downregulation of major pro-angiogenic factors, most notably VEGFA, in F3-KO GSCs compared to wildtype controls. Functionally, endothelial cells exposed to conditioned media from F3-KO GSCs exhibited a significantly reduced angiogenic capacity. This impairment was molecularly characterized by a distinct decrease in EdU-positive endothelial cells, indicating a robust suppression of endothelial cell proliferation and cell cycle progression in the absence of GSC-derived F3.

Conclusion: These results demonstrate that F3 plays a fundamental role in the basic mechanisms governing GSC-induced angiogenesis by regulating key pro-angiogenic players like VEGFA. Knockout of F3 downregulates crucial signaling pathways and directly impairs endothelial proliferation, highlighting F3 as a key molecular orchestrator of the vascular microenvironment in GB.



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